SPECIFICATION

Background of the Invention

High-cardinality parallel interfaces allow for the transmission of multi-byte data blocks in a single state of a single physical channel (e.g., one pixel). While this achieves a high level of data density, there is a need to encode even larger data blocks that may exceed the state-space of a single channel. For instance, a cryptographic key or a 64-bit data type is larger than the 24 bits of data that can be encoded in a standard pixel. A method is needed to structure data across multiple parallel channels in a way that further increases the effective data density of the physical layer.

Summary of the Invention

The present invention is a system and method for n-gram parallel data encoding, which increases the data density of a parallel communication interface by creating logical channels from groups of physical channels. An "n-gram" is a defined grouping of 'n' adjacent physical channels (e.g., pixels) that are treated by the encoding system as a single, unified logical channel.

The method involves defining the size of the n-gram (e.g., n=2 for a "bigram," n=3 for a "trigram"). The total number of unique states that this logical n-gram channel can represent is the product of the states of its constituent physical channels. For example, if a single pixel can represent N states, a bigram of two pixels can represent N*N = N$\hat{2}$ unique states. A communication system can then map a data block of a correspondingly larger size to a single state of the logical n-gram. This allows for the encoding of, for instance, a 6-byte (48-bit)

data block into the combined state of two adjacent 24-bit pixels. This technique achieves a form of data compression directly on the physical layer by maximizing the utilization of the parallel channel architecture.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the difference between single-channel encoding and n-gram encoding. **FIG. 2** is a block diagram showing the data flow for an n-gram encoding system.

Detailed Description of the Invention

Referring to **FIG. 1**, a conceptual comparison is shown. In a standard single-channel encoding scheme (100), a first data block (101) is mapped to the state of a first physical channel (103), and a second data block (102) is mapped to the state of an adjacent second physical channel (104). In the n-gram encoding method (110) of the present invention, a single, larger data block (111) is mapped to the combined state of a logical group (112) comprising both the first physical channel (103) and the second physical channel (104). This logical grouping is an n-gram, in this case a bigram.

FIG. 2 illustrates the data flow. A stream of data (201) is received by an n-gram encoder (202). The encoder groups the data into large data blocks (203) suitable for n-gram encoding. For each block, the encoder determines the required combined state for a group of 'n' physical channels (204) and instructs the transmitter (205) to set those channels to the appropriate states. On the receiving end, the sensor (206) reads the states of the group of channels, and an n-gram decoder (207) uses the combined state to reconstruct the original large data block

(208).